

Ayushmaan Singh

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[Leetcode](#) | [LinkedIn](#) | [GitHub](#):

EDUCATION

Vellore Institute of Technology

MCA (Expected June 2027)

CGPA: 8.58

Relevant Coursework:

Machine Learning, Data Science, Advanced Java, MySQL

Techno India Kolkata

Bachelor of Computer Applications (2022–2025)

CGPA: 7.96

Relevant Coursework:

DSA, OOPs, DBMS, Operating Systems, HTM, CSS, JS, Shell Scripting

TECHNICAL SKILLS

Languages: **Java, Python, SQL**

Cloud Tools: **AWS S3, AWS Lambda, Firebase**

Backend: **SpringBoot, FastAPI**

GenAI: **RAG Development and LLM Integration**

Database: **MySQL, MongoDB**

Tools: **Git, Docker, Codex**

PROJECTS

Minit – Quick Commerce Delivery

Node-JS | Kafka | Redis | MongoDB

- Developed a Real-Time quick commerce delivery system, with asynchronous messaging.
- Implemented Caching with Redis – for product catalog and current inventory.
- Utilized Apache Kafka as a distributed event bus to enable asynchronous communication between services to achieve high throughput and data consistency.

LearneXa - AI Enabled E-Learning Platform

Spring AI | Spring Boot | React and Material UI | MySQL | PGVector

- Integrated Spring AI to develop a Retrieval-Augmented Generation (RAG) system, allowing students to perform semantic queries across uploaded course documents.
- Utilized the Gemini API via Spring AI to automate the generation of certification quizzes and assignments, reducing manual content creation time by 85%.
- Leveraged **Inngest's** built-in observability to monitor async job execution, retries, and failures for reliable AI-generated content delivery.

AutoXP – AI-Powered Used Car Price Prediction & Marketplace

FastAPI | React | MongoDB | Scikit-Learn

- Engineered a full-stack vehicle valuation platform that predicts fair market prices using 10+ vehicle attributes, achieving 90%+ prediction accuracy.
 - Built an end-to-end machine learning pipeline involving data preprocessing, feature engineering, and ensemble regression models.
 - Developed and optimized machine learning models using Random Forest and XGBoost regression algorithms, leveraging feature selection, categorical encoding, and cross-validation techniques.
 - Implemented a health-check endpoint verifying ML model availability and database connectivity, and configured automated uptime monitoring with failure alerting on the live-hosted service.
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INTEREST

Data Driven Learning: Building prediction and recommendation engines using real-world

dataset Problem Solving: Solved 120+ DSA problems on LeetCode.